

Beyond Catalogue Counts: the Dataset Visibility Asymmetry in Low-Resource Multilingual NLP

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Slides, repo, and dataset
search



Before we label a language "low-resource", we need to ask what kind of shortage we mean.

**"Low-resource" may mean lacking resources.
It may mean resources are not visible.**

The distinction matters because labels shape benchmarks, shared tasks, funding priorities, and where new resource work is directed.

The question is simple: are resources truly absent, or are they missing from the places where counting happens?

"Low-resource" has infrastructural and social roots

Nigatu et al. (EMNLP 2024) point to three dimensions.

Most directly measured

▮▮ Technological infrastructure

Datasets, tools, models, and compute resources form the infrastructure a language can draw on for NLP.

≡ Linguistic & social status

Complex linguistic features, limited standardisation or digitisation, limited mainstream use, and curation constraints.

●●● Human & community resources

Limited native-speaker and linguistic/NLP expertise, plus limited capacity or willingness to provide data.

So we ask: when a language looks low-resource in datasets, does it truly lack datasets, or are they simply harder to see and find?

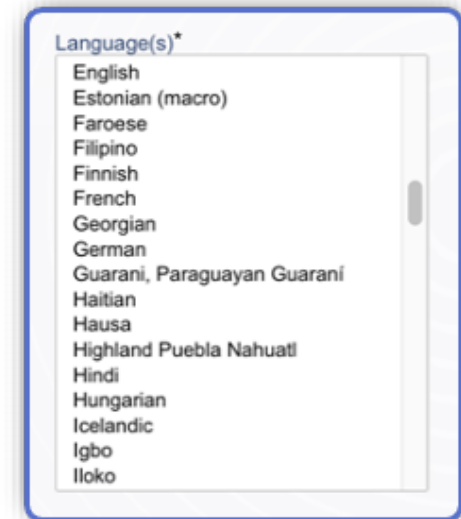
LRE Map and LDC make dataset counts easy to read

Both are central catalogues for language resources, and both expose language-level search.

LRE Map



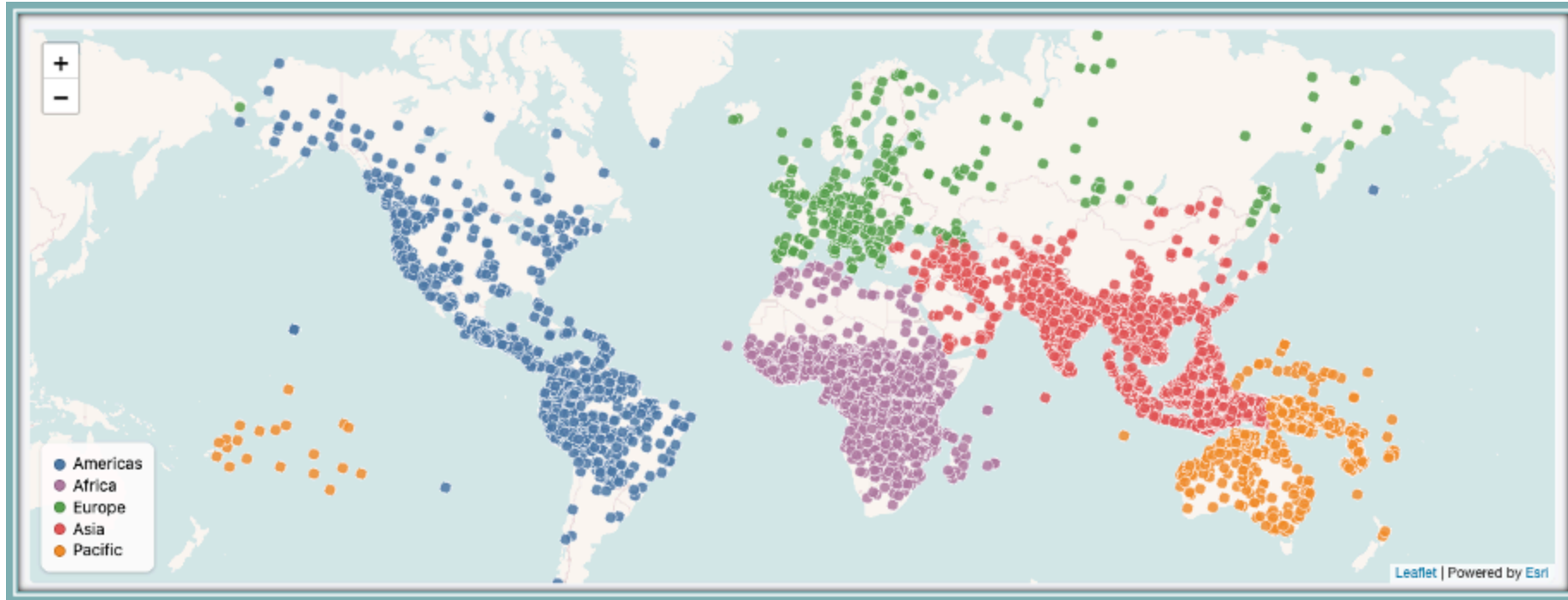
LDC catalogue



Two collection modes: LRE Map relies on authors submitting dataset metadata; LDC actively collects and distributes resources for a fee.

We examine the **200 most widely spoken** languages

Ethnologue lists 7,170 languages in use in the 2025 edition.



We set the scope to **Ethnologue 200** and ask, language by language, how many dataset records LRE Map and LDC show.

Large speaker communities, uneven catalogue visibility

Top-10 and bottom-10 of Ethnologue 200.

 LRE Map  LDC

English outlier compressed; others linear

Top of the list · biggest speaker communities

LANGUAGE	POP. M	CATALOGUE RECORDS	
English	1,500.0	3574 	640 
Mandarin Chinese	1,200.0	255 	224 
Hindi	609.1	104 	27 
Spanish	558.5	376 	103 
Standard Arabic	334.8	119 	119 
French	311.9	517 	34 
Bengali	284.3	53 	14 
Portuguese	266.6	203 	17 
Russian	253.4	161 	36 
Indonesian	252.4	31 	3 

Bottom of the list · still ~5 million speakers each

LANGUAGE	POP. M	CATALOGUE RECORDS
Pular	4.9	0
Tatar	4.8	0
Awadhi	4.8	0
Lao	4.7	7
Sicilian	4.7	0
Southern Uzbek	4.7	0
Luba-Katanga	4.7	0
Shan	4.7	0
Swati	4.7	0
Southern Thai	4.5	0

The bottom-10 languages have nearly zero datasets in the catalogues. The top-10 languages show substantial dataset counts.

From Resource Count to Resource Realization Density

Speaker-base-adjusted Resource Density Index (RDI)

$$RDI_i = \frac{\text{Datasets}_i}{\text{Speakers}_i / 1,000,000}$$

← **Dataset count:** realized resource supply

← **Speaker population:** scale of potential resource base

Other factors: conditions of resource realization
socioeconomic context, NLP expertise, language difficulty, etc.

Top of the list · biggest speaker communities

LANGUAGE	POP. M	CATALOGUE RECORDS	CATALOGUE RDI
English	1,500.0	3574  640	1.40
Mandarin Chinese	1,200.0	255  224	0.20
Hindi	609.1	104  27	0.11
Spanish	558.5	376  103	0.43
Standard Arabic	334.8	119  119	0.36
French	311.9	517  34	0.88
Bengali	284.3	53  14	0.12
Portuguese	266.6	203  17	0.41
Russian	253.4	161  36	0.39
Indonesian	252.4	31  3	0.07

Beyond Catalogue Counts

Catalogue RDI = $(RDI_{LRE\ Map} + RDI_{LDC}) / 2$

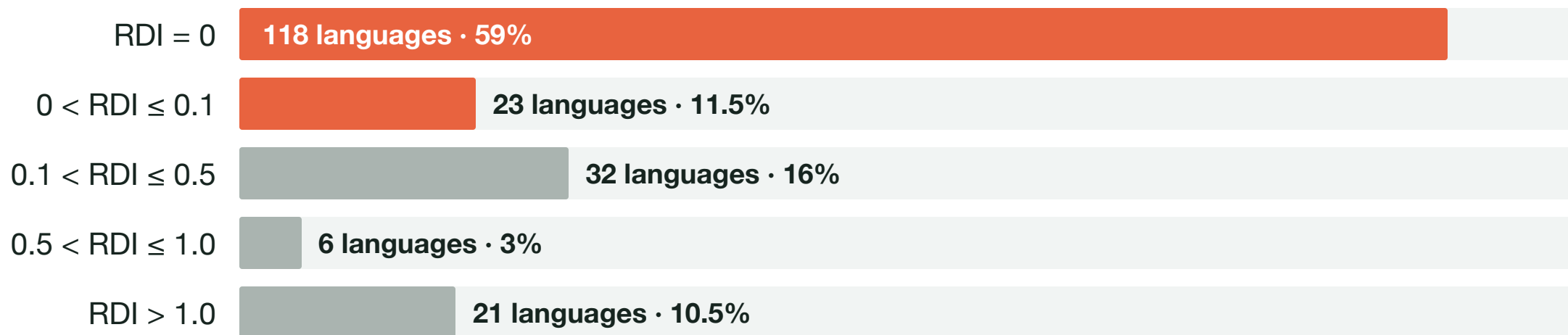
How to read it?

RDI = 0 No counted dataset from this evidence source.

RDI ≤ 0.1 At most 1 counted dataset per 10M speakers.

What does the catalogue view actually look like?

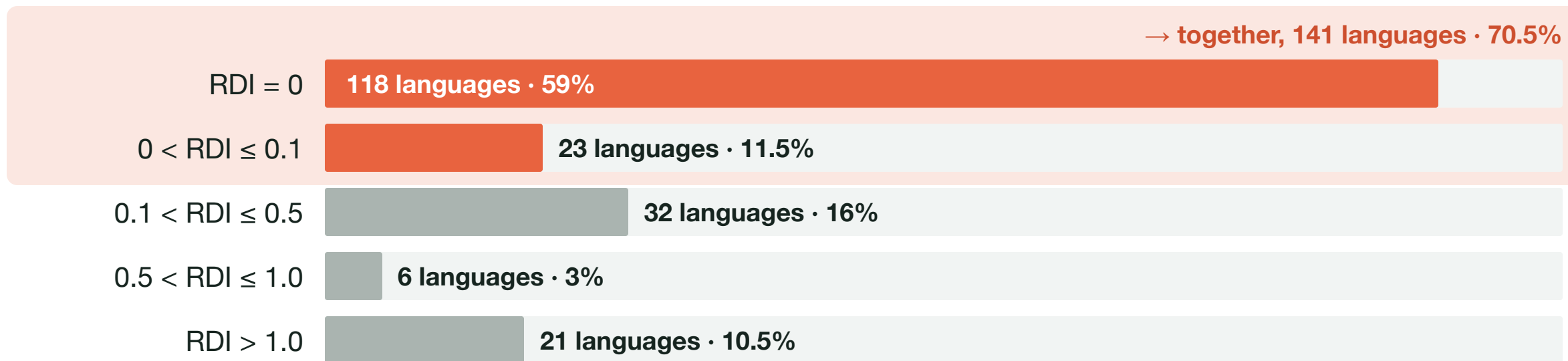
Average catalogue RDI (LRE Map + LDC) for the 200 most widely spoken languages.



Heavily skewed toward zero. **Median RDI = 0.0.** About 21 languages hold most of the catalogued datasets. The rest are effectively absent from both.

141 of 200 languages are at RDI 0.1 or below

At most **one** catalogued dataset per **ten million** speakers.



118 languages with nothing in either catalogue, plus **23** more below RDI 0.1. Together, they are **141** languages the catalogue view calls "severely under-resourced". **These are the 141 we examine next.**

The catalogue baseline makes these 141 languages look nearly absent.

**But low catalogue visibility
does not mean "no datasets".**

A dataset can be **created, cited, and re-used** in research, or shared through Hugging Face, OpenSLR, institutional, or regional repositories, without appearing in LRE Map or LDC.

We therefore look where dataset use leaves a trace in the research literature.

Datasets leave traces in the papers that use them

When catalogues miss or under-record a dataset, citations can still make it visible.



Read enough citation contexts and we can rebuild a **language-by-language repository** of datasets in research circulation, catalogued or not.

For the **141 low-visibility languages**, a small number of additional datasets can shift the measured visibility.

Scaling the search requires a pipeline

M3D (Tan & Duan, JCDL 2025) is our starting point for finding dataset evidence in citation text.

Why a pipeline?

Dataset evidence is embedded inside papers, often in citation contexts. Across **141 languages**, we need a repeatable way to search broadly before review.

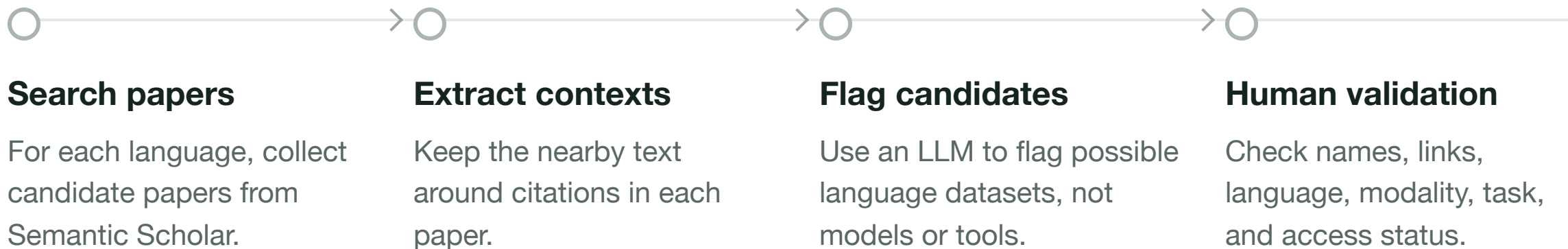
What changes here?

M3D was built to find datasets for a research question. Here we repurpose it to **study dataset visibility language by language**, then turn candidates into checked dataset records.

For each language, the pipeline lets us automatically collect dataset evidence from papers at scale.



From papers to a checked dataset repository

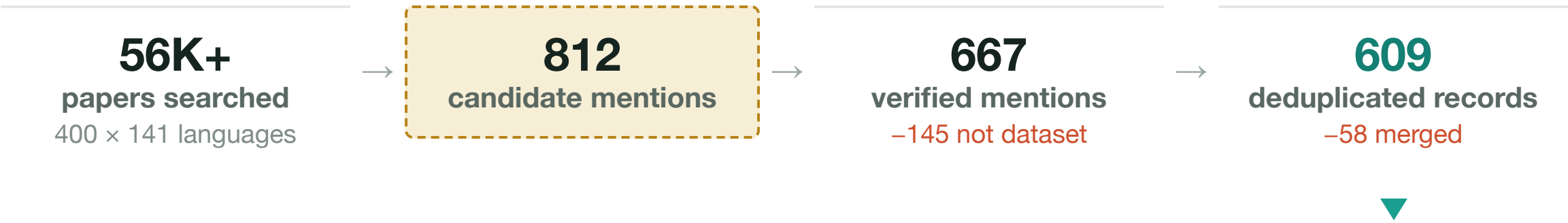


The output is a **checked repository of literature-evidenced datasets** for the 141 low-visibility languages.



Validation turns 812 mentions into 609 dataset records

Paper search produces candidate mentions. Validation and merging produce dataset records.



One deduplicated dataset record

Dataset name

LangText Corpus

Language

Lang.

Dataset paper

Doe 2020

Modality

text

Dataset URL

repo.org/langtext

Citing paper

Smith 2024

Task

classification

Access

available

What changes when we add literature evidence?

We start with a few examples, then zoom out to all 141 languages.

Nepali	0	→	30	not in any catalogue · RDI 0.0 → 0.9
Setswana	0	→	26	not in any catalogue · RDI 0.0 → 1.9
Assamese	1	→	31	1 LDC entry · RDI 0.02 → 1.31
Marathi	14	→	41	14 in LRE Map · RDI 0.07 → 0.41
Indonesian	34	→	196	community NusaCrowd already lists 137 · RDI 0.07 → 0.78



Across all 141 languages, 53 surfaced in papers

The full view shows where literature evidence appears and where it does not.

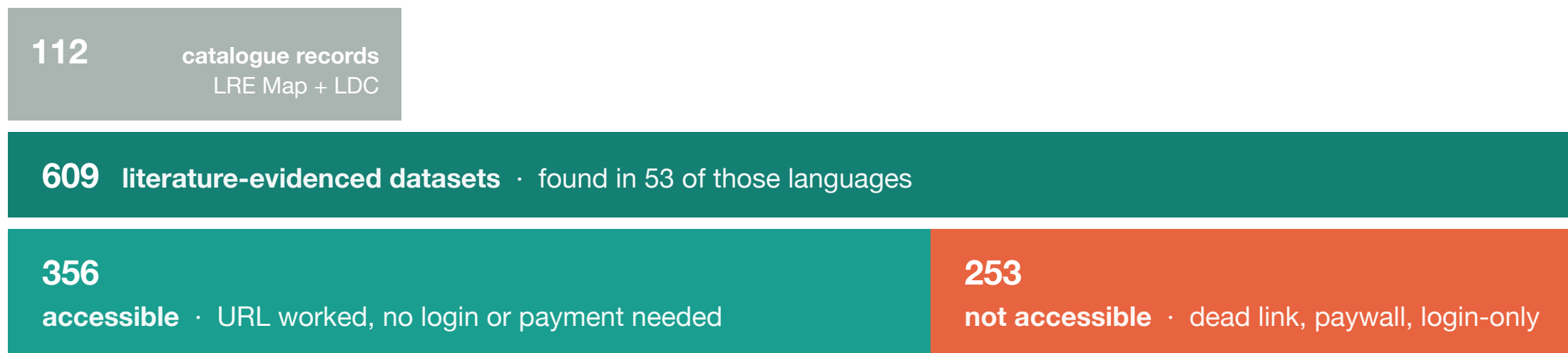
RDI = 0 ■ 35 literature only ■ 83 neither

$0 < \text{RDI} \leq 0.1$ ■ 18 both ■ 5 catalogue only

apd	bar	bho	ctg	dje	hae	kir	lin	luo	mag	mai	mya	nod	npi	nso	nya	ory	pan	pbu	pcm
pnb	pst	run	sck	sdh	snd	sot	syl	tat	tir	tsn	tts	tuk	wes	xho	acq	acw	aec	ars	awa
ayh	ayl	ayn	azb	azj	bam	bci	bgc	bjj	bns	cjy	czh	dcc	dyu	ewe	fub	fuc	fuf	fuv	gan
gax	gaz	gpe	haz	hil	hne	hnm	hsn	ibb	ibo	ilo	ins	kab	kam	kas	kik	knc	kng	kri	lir
lmn	lua	lub	lug	mad	mey	min	mkw	mnp	mos	mup	mwr	nap	pbt	plt	rkt	sag	sat	scn	shi
shn	shu	sid	skr	sna	sou	ssw	suk	sun	swc	tgk	tiv	tso	umb	uzn	uzs	vah	vjk	aka	asm
ceb	ckb	guj	hau	ind	jav	khm	kin	kmr	mar	nan	som	swh	uig	wol	yor	ayp	cdo	gug	hak
wuu																			

Not surfaced in papers means **not found by this search**, not proof that no datasets exist.

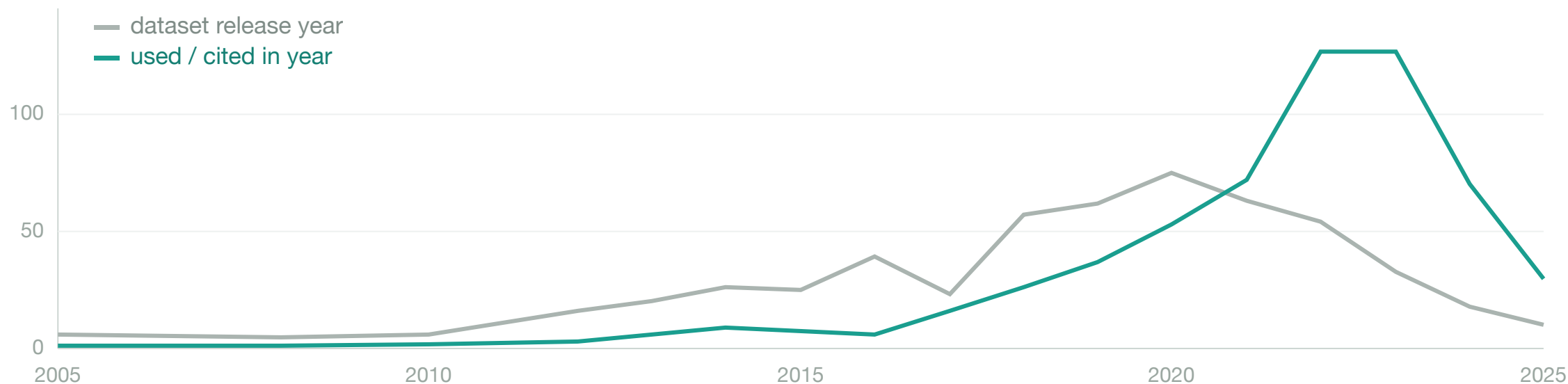
Even reachable datasets outnumber catalogue records



The good news: **our method works**. The bad news: being used and verified in papers does not guarantee that every dataset remains reusable.

Limitation: citation evidence has timing limits

No citation means no signal, and new datasets need time to appear in citing papers.



Dataset releases climb through the 2010s and peak around 2020. Usage peaks in 2022–2023. The late dip likely reflects **citation latency**, because recent datasets have had less time to be cited.

Limitation: what we still miss

These gaps mostly make the count smaller, not larger.

- **No citation, no signal** means used-but-not-cited datasets stay invisible.
- **Citation latency** means recent datasets may not have enough time to surface in citing papers.
- **English and access bias** means local-language papers, paywalled venues, and national repositories can be missed.
- **Search limits** mean the top-400 paper cutoff and the search queries may miss relevant papers.
- **LRE Map + LDC baseline** does not include Hugging Face Hub, OpenSLR, or regional registries.

Read the **609 records** as evidence that at least this much is already visible in papers.

Make visibility easier to check, update, and credit

Incomplete evidence can make languages look data-poor and steer benchmarks, funding, and inequality narratives. The problem is not only finding more datasets. It is keeping evidence usable for everyone.

Check

Transparent search

Publish queries, source papers, dataset records, and uncertainty so others can inspect the path from paper to count.

Update

Living records

Let communities add datasets, fix URLs, flag access changes, and link related records without restarting the whole search.

Credit

Shared incentives

Make dataset creators, local-language curators, catalogue maintainers, and repository contributors visible in the same record.

Released openly, built to grow



Slides, search system,
data, statistics, code

The repo includes the RDI table, the 609-dataset repository with source and citing papers, access status and URLs, and the full pipeline.

- **Use it.** Search the records, links, statistics, and claims.
- **Extend it.** Run the method for non-English literature, sub-domains, models, and tools.
- **Add to it.** Add a dataset, fix a URL, open an issue, or send a pull request.

Thank you. · github.com/zhiyintan/dataset-visibility-asymmetry